

Tutorial 1 Week 3: Real World Planning and Acting

Tutorial Team 26: Chao Chien Hung, Yao Tianhao

August 31, 2026

1 Problem 1 – A Blocks World

The initial state is a single tower $A-B-C$, with A at the top and C on the table. The goal is to put B on the table and leave A on C .

(a) PDDL formulation

The table is treated as having unlimited clear space. Therefore moving a block onto the table does not make the table non-clear.

```
(define (domain middle-pop)
  (:requirements :strips :typing :equality)
  (:types block)
  (:predicates
    (on ?x - block ?y - object)
    (clear ?x - object))

  (:action move-to-block
    :parameters (?x - block ?from - object ?to - block)
    :precondition (and (on ?x ?from)
                       (clear ?x)
                       (clear ?to)
                       (not (= ?x ?to)))
    :effect (and (not (on ?x ?from))
                 (on ?x ?to)
                 (clear ?from)
                 (not (clear ?to))))

  (:action move-to-table
    :parameters (?x - block ?from - object)
    :precondition (and (on ?x ?from)
                       (clear ?x)
                       (not (= ?from table)))
    :effect (and (not (on ?x ?from))
                 (on ?x table)))
```

```

(clear ?from)))

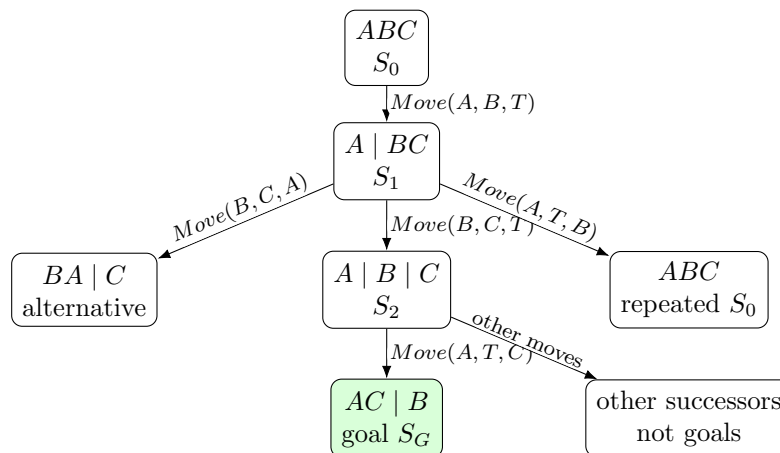
(define (problem middle-pop-instance)
  (:domain middle-pop)
  (:objects a b c - block table - object)
  (:init (on a b) (on b c) (on c table)
         (clear a) (clear table))
  (:goal (and (on a c) (on b table) (on c table))))

```

All atoms omitted from a state are false under the closed-world assumption.

(b) Forward-search tree

Write ABC for a tower in which A is above B , which is above C , and use “|” to separate stacks on the table. The useful part of the forward-search tree is:



Initially only A is clear, so it must first be moved to the table. This makes B clear. Moving B to the table makes C clear, after which A can be placed on C .

(c) Valid plan

The following is a shortest valid plan:

$\langle \text{Move}(A, B, \text{Table}), \text{Move}(B, C, \text{Table}), \text{Move}(A, \text{Table}, C) \rangle$.

2 Problem 2 – Heuristic for Planning

When preconditions and goals contain only positive literals, removing all negative (delete) effects makes facts accumulate monotonically. After every action, the relaxed state contains a superset of the facts in the original state. Hence every action applicable in an original plan remains applicable in the relaxed problem, and every positive goal achieved in the original problem is also achieved in the relaxation. The relaxed problem therefore cannot be harder, and its optimal cost is a lower bound on the original optimal cost.

This no longer necessarily holds if negative preconditions or goals are allowed. For example:

- If initially p is true, action a deletes p , and the goal is $\neg p$, then $[a]$ solves the original problem. Removing the delete effect makes the goal unreachable.
- If initially p is true, action a deletes p , and action b has precondition $\neg p$ and effect g , then $[a, b]$ reaches g in the original problem. In the relaxed problem p remains true, so b cannot be executed.

Therefore delete removal is guaranteed to be a relaxation only under the stated positive-precondition and positive-goal restriction.

3 Problem 3 – Hierarchical Task Planning

(A) HLA schemas

1. Both implementations of H_1 first use P_1 to make A true. The final action either adds B or deletes B . Thus

$$\text{Action}(H_1, \text{Precond} : \neg A, \text{Effect} : A \wedge \sim \pm B),$$

with implementations $[P_1, P_2]$ and $[P_1, P_3]$.

2. Every implementation of H_2 requires $\neg B$. Some implementations add A (none deletes it), while C may be added or deleted. Thus

$$\text{Action}(H_2, \text{Precond} : \neg B, \text{Effect} : \sim +A \wedge \sim \pm C),$$

with implementations $[P_4]$, $[P_5]$, $[P_6]$, and $[P_7]$.

3. P_8 requires $\neg B \wedge \neg C$ and establishes D . Then either P_9 or P_{10} establishes E . Thus

$$\text{Action}(H_3, \text{Precond} : \neg B \wedge \neg C, \text{Effect} : D \wedge E),$$

with implementations $[P_8, P_9]$ and $[P_8, P_{10}]$.

(B) Which sequences achieve E ?

Under the closed-world assumption, the initial state $\neg A \wedge B$ also implies $\neg C$, $\neg D$, and $\neg E$.

- (i) **Valid.** $[P_1, P_3]$ establishes $A \wedge \neg B$; P_6 establishes $\neg C$; $[P_8, P_9]$ then establishes D and E .
- (ii) **Valid.** $[P_1, P_3]$ establishes $\neg B$ and P_7 establishes $\neg C$. Then P_8 establishes D , allowing P_{10} to establish E .
- (iii) **Invalid.** $[P_1, P_2]$ makes B true, but P_4 requires $\neg B$.
- (iv) **Invalid.** $[P_1, P_2]$ makes B true, but P_5 requires $\neg B$.

Therefore the correct choices are

(i) and (ii).